

TABLE II
Test Points

Test point	Details
TP0	GND
TP1	5V
TP2	Logic Input for motor M1
TP3	Logic Input for motor M1
TP4	Logic Input for motor M2
TP5	Logic Input for motor M2
LED1 button pressed	LED1 blinks
LED2 button pressed	LED2 blinks

Solution Explorer window are correct as shown in Fig. 5.

If any of the references is not there, right-click 'References' to add it from .NET section. While building the solution, keep 'Copy Local' under 'Properties' as 'True' for 'Coding4Fun.Kinect.Wpf', 'Microsoft.Research.Kinect' and 'Microsoft.Speech' so that these files are copied to the release folder also for convenience. 'Path' property of 'Coding4Fun.Kinect.Wpf' should point to the corresponding file in the release folder for error-free build.

BASCOM program. The software for the robot is written in BASIC language and compiled using BASCOM compiler. The program keeps comparing the commands received from the serial port with 's', 'w', 'v', 't', 'l', 'a' and 'b', and initiates corresponding processes as shown in Table I.

Construction and testing

Interface Kinect sensor. In order to set up Kinect sensor and interface it to the computer, below-mentioned software need to be installed before connecting Kinect sensor to the computer:

1. Microsoft Visual Studio 2010 Express or other Visual Studio 2010 edition
2. .NET Framework 4 (installed with Visual Studio 2010)
3. DirectX Software Development Kit, June 2010 or later version
4. DirectX End-User Runtime Web Installer
5. Kinect Software Development Kit - v1.0 - beta2 - x86
6. Microsoft Speech Platform - Server Runtime, version 10.2 (x86

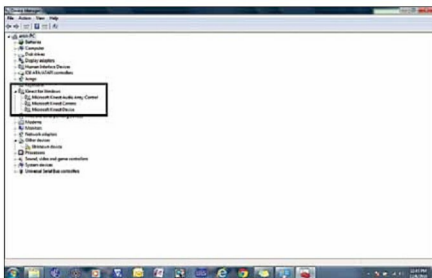


Fig. 6: Correct detection and installation of Kinect sensor

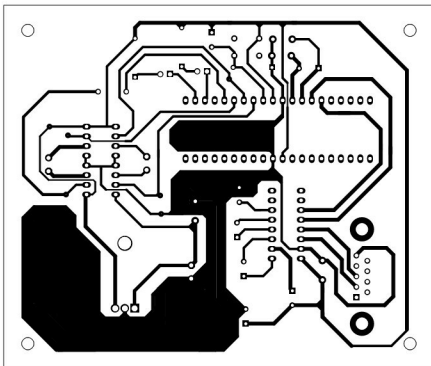


Fig. 7: An actual-size, single-side PCB for the robot

edition)

7. Microsoft Speech Platform - Software Development Kit, version 10.2 (x86 edition)

8. Kinect for Windows Runtime Language Pack, version 0.9 (acoustic model from Microsoft Speech Platform)

The software versions need to be strictly respected for correct operation of the project.

Once everything is installed, connect Kinect sensor to the computer's USB port. It will be automatically detected and drivers installed. The correct detection and installation of the device is shown in Fig. 6. Kinect takes up COM port 3 by default.

Interface robot to the computer. An actual-size, single-side PCB layout of the robot is shown in Fig. 7 and its component layout in Fig. 8. Assemble